



Using a Venn diagram to display outcomes for two events

Task A: Venn diagrams

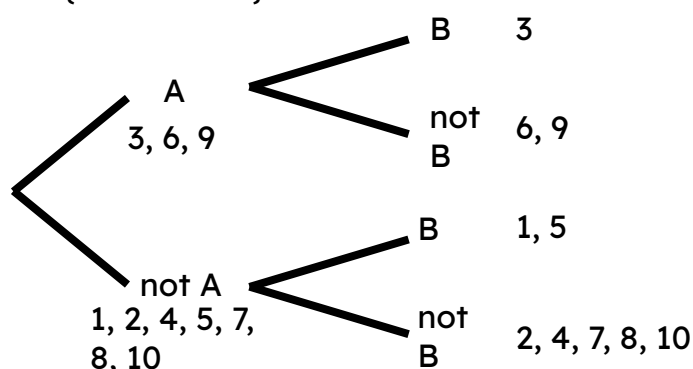
1) Here is an outcome tree for two events.

Draw a **Venn diagram** to show the same information.

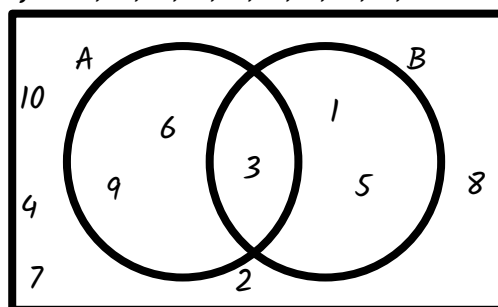
The two events are:

$A = \{\text{multiple of } 3\}$

$B = \{\text{factor of } 15\}$



$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

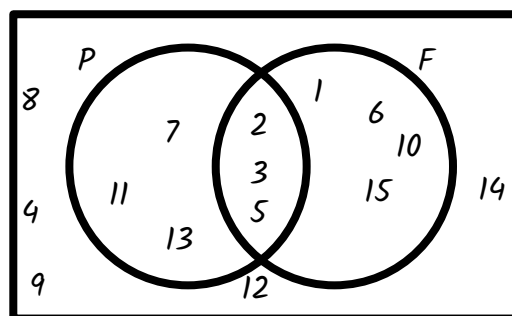


2) Here is an outcome table for two events.

Draw a **Venn diagram** to show the same information.

	prime number (P)	not a prime (P')
factor of 30 (F)	2 3 5	1 6 10 15
not a factor of 30 (F')	7 11 13	4 8 9 12 14

$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

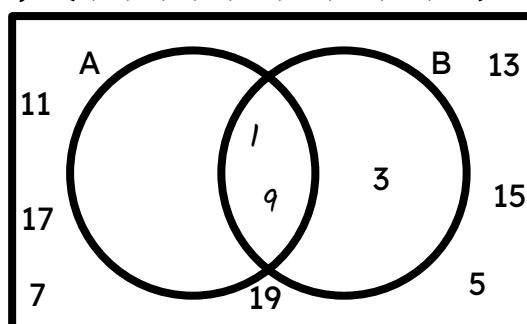


3) Here is a partially complete outcome table and **Venn diagram**.

Complete both diagrams.

	square number (A)	not a square number (A')
factor of 18 (B)	1 9	3
not a factor of 18 (B')		5 7 11 13 15 17 19

$\xi = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$



Task B: Outcomes from Venn diagrams

1) Use this **Venn diagram** to answer the following questions:

a) List the outcomes that are in event A.

$\{1, 2, 3, 7, 8, 12, 13, 15\}$

b) List the outcomes that are in event B.

$\{2, 4, 5, 8, 9, 14, 15\}$

c) List the outcomes that are in both event A and event B.

$\{2, 8, 15\}$

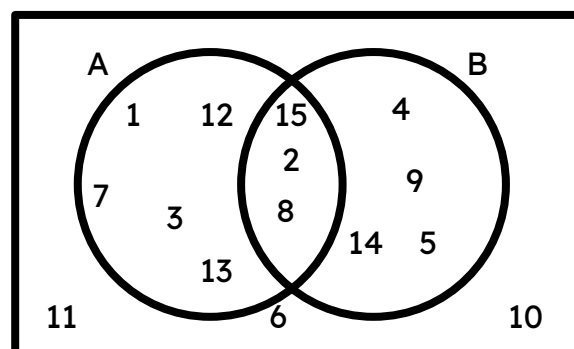
d) List the outcomes that are in event A but not in event B.

$\{1, 3, 7, 12, 13\}$

e) List the outcomes that are not in either event A nor event B.

$\{6, 10, 11\}$

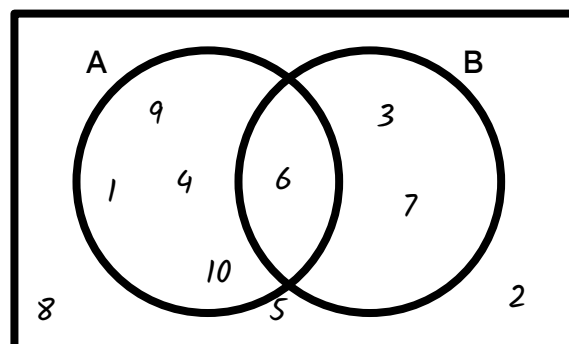
$\xi = \{\text{integer values from 1 to 15 inclusive}\}$



2) a) Use the following information to complete the **Venn diagram**.

- the possible outcomes from the trial are the integers from 1 to 10 inclusive
- there are five possible outcomes for event A
- $\{6\}$ is the only outcome of both events
- $\{2, 5, 8\}$ are not outcomes of event A nor event B
- the outcomes that are only in event B are prime

$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$



b) List the outcomes of event A.

$\{1, 4, 6, 9, 10\}$

c) List all the outcomes for event A or event B or both.

$\{1, 3, 4, 6, 7, 9, 10\}$

3) Use this **Venn diagram** to answer the following questions:

a) How many outcomes are in the sample space?

12 outcomes

b) How many outcomes are in event A?

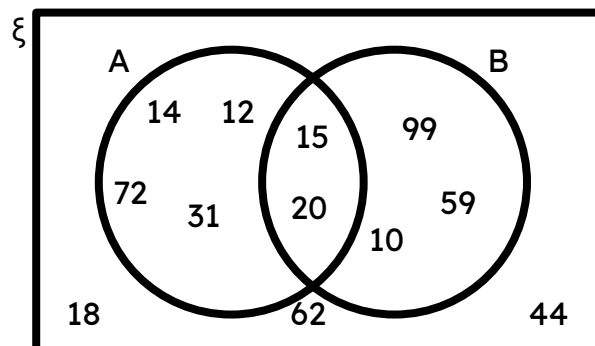
6 outcomes

c) How many outcomes are in event B?

5 outcomes

d) How many outcomes are not in event A or event B?

3 outcomes



e) Explain why your answers to parts b, c and d do not sum to part a.

There are two outcomes that exist in both event A and event B so are counted twice in the sum.

